

$$C_{\phi W}^{(11)} \rightarrow$$

$$\begin{aligned} & \frac{1}{16 \pi^2} \left(\frac{1}{16} \sum_{\mathbf{p}} g_1^2 g_2^2 \text{LF}_{3,0}[\mathbf{m}_{\tilde{l}}^{\mathbf{p}}] - \frac{1}{16} \sum_{\mathbf{p}} g_1^2 g_2^2 \text{LF}_{4,-1}[\mathbf{m}_{\tilde{l}}^{\mathbf{p}}] - \frac{1}{16} g_2^2 (6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_{\mathbf{p}} g_1^2) \text{LF}_{3,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \right. \\ & \frac{1}{16} g_2^2 (6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_{\mathbf{p}} g_1^2) \text{LF}_{4,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \frac{5}{4} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] + g_2^4 \text{LF}_{3,1,-1}[\mathbf{m}_2, \tilde{\mu}] - \\ & g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 \text{LF}_{4,1,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{l}}^{\mathbf{p}}, \mathbf{m}_{\tilde{e}}^{\mathbf{r}}] + \\ & \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{l}}^{\mathbf{p}}, \mathbf{m}_{\tilde{e}}^{\mathbf{r}}] - \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{d}}^{\mathbf{r}}] + \\ & \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{d}}^{\mathbf{r}}] - \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] - \\ & \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \\ & \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \\ & \frac{3}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] - \frac{9}{8} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \\ & \frac{3}{4} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] - \frac{1}{8} g_1^2 g_2^2 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_1] + \\ & \frac{3}{8} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{4} g_1^2 g_2^2 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \frac{3}{8} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] - \\ & \left. g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \frac{9}{8} g_2^4 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{4} g_2^4 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_2] \right) \end{aligned}$$